

ALGORITHM

- 1) Start
- 2) Prompt the user for vehicle type, current battery level and availability. Store each data in variable.
- 3) If there is no availability
 - If the vehicle type is Hybrid
 - Display that no charging, parking only
 - Stop
 - Else Display that no charging and stop
- Else
 - If the vehicle type is Hybrid
 - if current battery level is greater than 41
 - * Display "not qualified"
 - * stop
- 4) Prompt the user to enter a target battery level and store it.
- 5) Set units to (target - current) and display that there's no need for charging if units is negative or zero.
- 6) Prompt the user for time in 24-hour format, expected operation, membership and disability status.
- 7) If current battery level is less than 16 and target battery level is greater than 79 Then Set prt ← 'E'
 Else if the customer is disabled or has a membership and has a current battery level of 30 or less then prt ← 'V'
 Else set prt ← 'N'

8) If time is in between 17 and 22 (inclusive)

- set season $\leftarrow$ 'P'

- set Crate $\leftarrow$ 50

- If priority is not emergency and membership is available then set Charging Discout $\leftarrow$ 10

Else

- set season $\leftarrow$ 'O'

- set Crate $\leftarrow$ 35

- If priority is not emergency and membership is available then set parking discount $\leftarrow$ 20
Charging

a) If customer is not disabled

- If duration is 2 or less then set Parkingrate $\leftarrow$ 200

- If duration is between 2 and 5 (exclusive) then set parking discount $\leftarrow$ 400
rate

- If duration is 5 or more then set parking rate to 700

- If membership exists then set parking discount $\leftarrow$ 20

Else set prate $\leftarrow$ 0 and pdiscout $\leftarrow$ 100

10) Set electric cost to units times charging rate with discount

11) Set parking cost to parking rate with discount

12) Set final amount to sum of e-cost and P-cost.

- 13) If priority is emergency then
 ■ Display "emergency priority"
Else if priority is very important then
 ■ Display "Priority customer"
Else display "Normal customer"
- 14) If season is peak then display :
 "Season : peak" else display "Season : off-peak"
- 15) Display current and target battery level
- 16) Display charging, parking and total cost.
- 17) Enter applied discount-
- 18) If duration is more than 8 then display
 a Long-stay warning message else display
 "Standard parking duration"
- 19) Stop.